FYP MEETING LOG

Supervisor Meeting 1 of 3

Agent-Based Game Analysis in Unity Based Upon Curiosity and RND Exploration Using PPO and SAC
Student: Raza Ullah Khan (NUM-BSCS23-28) | Supervisor: Dr. Malik M Ali Shahid | Co-Supervisor: Mr. Abdul Rafay | Industrial Partner: Inspired Square

Meeting No.	02 (Supervisor Meeting 1)
Meeting With	Supervisor - Dr. Malik M Ali Shahid
Date	Monday, 12 October 2026
Time	11:00 AM - 11:30 AM
Duration	30 minutes
Mode / Venue	Supervisor's office, Namal University Mianwali
Attendees	Raza Ullah Khan (Student); Dr. Malik M Ali Shahid (Supervisor)
Milestone	FN-01 (Month 1, Weeks 1-2): Project Initialization, Development Environment & Version Control

1. Purpose of the Meeting

To share what the industrial partner said in last week's meeting, lock the project scope, and agree on the eight-month plan and how we will meet from now on.

2. Agenda

1. Feedback from the Inspired Square meeting.
2. Final scope: six conditions, one main game, selected runs in a second game.
3. Rules that keep the comparison fair.
4. Milestone plan (FN-01 to FN-16) and meeting routine.

3. What We Discussed

- **Industry feedback.** I told sir about the play-tester idea and the warning about reward hacking. Sir agreed that watching the agent is important, and said videos will also be useful in the final presentation.
- **Scope is fixed.** PPO and SAC, each with baseline, Curiosity and RND. The full matrix runs in Hummingbirds. In the second game we only repeat the comparisons that turn out to be interesting, not all six.
- **Fair comparison.** Sir stressed that only the algorithm and the exploration method should change. The task, observations, actions, reward, training steps and the way we measure must stay the same for all runs.
- **Random seeds.** One run is not enough because RL results change from run to run. We agreed on 5 seeds per condition, which means 30 runs in the main game. We will report the average and the spread (standard deviation).
- **Order of work.** First PPO and SAC without any exploration bonus. Only after these baselines are stable do we add Curiosity and RND. Otherwise we will not know what caused a change.
- **Writing style.** Sir asked to keep claims neutral in the report: say what we observed, not what we hoped.

4. Small Implementation Part

We went through the milestone plan for the first two months and the folder layout I plan for the GitHub repository. Each meeting from now on will be linked to the current milestone.

ID	Weeks	Milestone
FN-01	1-2	Project initialization, development environment and version control

FN-02	3-4	Hummingbirds 3D flight area and flower setup
FN-03	5-6	Agent observations, continuous actions and keyboard testing
FN-04	7-8	Python connection and first PPO baseline training run

Table: first four of the sixteen two-week milestones (8 months = 32 weeks).

```
unity-rl-exploration-ppo-sac/      (planned repository layout)
|-- README.md               project summary, matrix, metrics, team
|-- requirements.txt        Python packages (ML-Agents trainer)
|-- docs/                   proposal, slides, meeting logs
|-- Packages/manifest.json  Unity packages incl. ML-Agents
|-- ProjectSettings/        Unity project settings
|-- config/                 training YAML files (added at FN-04)
`-- .gitignore              ignore Library/, Temp/, results/, wandb/
```

5. Decisions Taken

- Scope approved: 6 conditions in Hummingbirds, selected conditions in a second game.
- 5 random seeds per condition; results reported as mean and standard deviation.
- Baselines (PPO and SAC, no bonus) must be finished before any exploration method is added.
- Supervisor meeting once a week (about 30 minutes); technical check-ins with the co-supervisor (about 20 minutes).
- A meeting log will be written after every meeting and kept in the docs folder.

6. Action Items

#	Action Item	Responsible	Due Date
1	Set up Unity, Python and the ML-Agents trainer on the lab PC.	Raza	23 Oct 2026
2	Create the GitHub repository with main and develop branches.	Raza	13 Oct 2026
3	Collect the key papers for PPO, SAC, Curiosity and RND.	Raza	26 Oct 2026
4	Meet the co-supervisor about the technical setup.	Raza	19 Oct 2026

7. Next Meeting

Tuesday, 27 October 2026 - literature and evaluation metrics.